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HASH CHAIN OF BLOCKS OF PUBLIC KEYS FOR MESSAGES

Abstract

A block to be added to a hash chain is generated. The block is for a message to be sent and includes a link hash of a prior block previously added to the hash chain. The prior block includes a public key corresponding to a private key to be used to digitally sign the block. The block includes a payload hash of a payload of the message, and a next public key corresponding to a next private key to be used to digitally sign a next block to be added to the hash chain after the block. The next block is for a next message to be sent after the message. The block is digitally signed with the private key and added to the hash chain. The link hash links the block to the prior block in the hash chain. The message is sent.

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Background/Summary

BACKGROUND

[0001] Computing and other types of electronic devices are often communicatively connected to networks so that they can send messages to each other. Asymmetric cryptography can be used for message authentication. Specifically, a sending device has a cryptographic key pair made up of a private key and a corresponding public key. The sending device digitally signs a message with the private key before sending it to a receiving device. The receiving device uses the corresponding public key to authenticate that the message was signed by the private key and, assuming that the receiving device trusts that the public key is associated with the sending device, thus that the message was sent by the sending device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1A is a diagram of an example hash chain of blocks of one-time signature (OTS) public keys for authenticating messages.

[0003] FIG. 1B is a diagram of an example non-origin block of the hash chain of FIG. 1A.

[0004] FIG. 2 is a flowchart of an example method for sending messages that can be authenticated using the hash chain of FIG. 1A.

[0005] FIG. 3 is a flowchart of an example method for receiving and authenticating messages using the hash chain of FIG. 1A.

[0006] FIG. 4 is a diagram of an example non-origin block of the hash chain of FIG. 1A in an implementation in which OTS public keys for authenticating multiple messages are included in the block.

[0007] FIG. 5A is a diagram of an example implementation in which Merkle or other binary trees are used for authenticating messages.

[0008] FIG. 5B is a diagram of an example non-origin block of the hash chain of FIG. 1A in the implementation of FIG. 5A.

[0009] FIG. 6 is a diagram of an example system in which messages from a sending device to a receiving device can be authenticated using the hash chain of FIG. 1A.

DETAILED DESCRIPTION OF THE DRAWINGS

[0010] As noted in the background, asymmetric cryptography can be used for message authentication. Upon receiving a message from a sending device that has been digitally signed with a private key, a receiving device can use the corresponding public key to authenticate that the message was sent by the sending device if the receiving device trusts that the public key is associated with the sending device. Traditionally a sending device has used the same private key to digitally sign multiple messages sent to the same or different receiving devices, with the receiving devices then using the same corresponding public key to authenticate the messages. This means that each receiving device may have to receive the public key just once, and moreover may have to authenticate the public key just once in order to trust that the public key is associated with the sending device.

[0011] However, with the advent of quantum computing, existing asymmetric cryptography is susceptible to compromise. The usage of quantum computing can permit a nefarious party to determine the private key that a sending device uses to digitally sign messages, and therefore to subsequently masquerade as the sending device. Quantum computing attack-resistant asymmetric cryptography schemes have been developed. For instance, hash-based signature schemes, such as the Leighton-Micali one-time signature (LM-OTS), the Winternitz one-time signature (WOTS), and the WOTS+ schemes are resistant to quantum computing attacks and can be used for digital signature and authentication purposes.

[0012] A hallmark of such hash-based quantum computing attack-resistant cryptography schemes is

the usage of OTSs. In such schemes, the private key of a given cryptographic key pair can be used just once, to digitally sign a single message. Therefore, the corresponding public key of the key pair is also used just once, to authenticate a single digitally signed message. So long as the private key is not used to digitally sign more than one message, quantum computing may not be able to be used to determine the private key. The private key is thus an OTS private key—and its corresponding public key is likewise an OTS public key—in order to maintain resistance to computing attacks, including quantum computing attacks.

[0013] A difficulty with using OTS public keys to authenticate messages is that the receiving device has to trust each public key is associated with the sending device in order to authenticate that the corresponding message was indeed sent by the sending device. It is impracticable for the receiving device to individually authenticate each public key, however. The Leighton-Micali signature (LMS) scheme, which uses LM-OTSs, and the extended Merkle signature scheme (XMSS), which uses WOTS+, provide ways to avoid a receiving device having to individually authenticate each OTS public key in order to trust that the public key is associated with a sending device.

[0014] For instance, $2^{\text{sup.h}}$ OTS private keys may be generated using a seed value and a state, such as a counter that is incremented from 0 to $2^{\text{sup.h}}-1$. Corresponding OTS public keys are generated, and combined in a binary tree known as a Merkle tree, which has height h . The OTS public keys correspond to the leaf nodes of the tree, and a root public key corresponds to the root node.

[0015] To authenticate that a message has been sent by the sending device, the receiving device does not have to receive the corresponding OTS public key, and thus does not have to authenticate that the OTS public key is itself associated with the sending device. Rather, the receiving device just has to authenticate that the root public key is associated with the sending device.

[0016] Specifically, the receiving device authenticates the message using the root public key, the digital signature of the message, and the authentication path from the leaf node corresponding to the OTS public key to the root node in the Merkle tree (where the receiving device has already authenticated the root public key as being associated with the sending device). The authentication path can be included in the message itself.

[0017] That is, the receiving device itself calculates a candidate OTS public key from the message and the digital signature of the message, the latter which the sending device generated when digitally signing the message with the OTS private key. The receiving device then verifies that the candidate OTS public key it calculated is part of the binary tree including the root public key, using the authentication path.

[0018] Such approaches permit authentication of up to $2^{\text{sup.h}}$ messages using the same trusted root public key. The value for $2^{\text{sup.h}}$ has to be decided before the root public key is generated, however, and cannot be subsequently changed. If the selected value for $2^{\text{sup.h}}$ is too small, a sending device may run out of OTS private keys. The OTS private keys and their corresponding OTS public keys have to be generated before the root public key can be generated, and thus before the first message is sent. Generation of the Merkle tree is therefore computationally expensive, and can be computationally prohibitive for lightweight devices such as Internet of Things (IoT) devices.

[0019] Selecting a large value for $2^{\text{sup.h}}$ to ensure that a sending device does not run out of OTS private keys is not a panacea, however. Merkle trees for large values of $2^{\text{sup.h}}$ can require a large amount of storage space, which lightweight devices in particular may not have. For example, schemes such as the LMS-hierarchical signature system (HSS) scheme and the multi-tree XMSS (XMSS $\{\text{circumflex over ()}\}$ MT) scheme, among others, permit the OTS public keys to not have to be stored after generation, but at the cost of requiring more computation later. Ultimately, quantum computing attack-resistant authentication has proven to either be computationally expensive and thus slow, require a relatively large amount of storage space such as volatile as well as non-volatile memory, or both.

[0020] Techniques described herein ameliorate these issues. The techniques employ a hash chain of blocks of OTS public keys. The origin, or root, block of the hash chain includes the OTS public key for the first message that will be sent by a sending device, even before the first message has been generated. After generating the first message but before sending it, the sending device adds a block to the hash chain for the message which links to the origin block. The block includes a payload hash of the message's payload, and is digitally signed with the OTS private key corresponding to the OTS public key included in the origin block. The block also includes the OTS public key that will be used to digitally sign the block for the next (i.e., second) message, which may not yet have been generated.

[0021] A receiving device just has to authenticate the origin block to trust that the subsequently sent messages digitally signed with OTS public keys in the hash chain were indeed sent by the sending device—and just has to authenticate the origin block once. The receiving device authenticates a message using the OTS public key included in the prior block that the block for the message links to. The sending device only has to generate one OTS public key in advance. When a new message N is to be sent, only at that time does the sending device have to generate another OTS public key. The newly generated OTS public key is for the next message N+1 and is included in the block N added for message N to the hash chain. The block N is linked to the previously added, prior block N-1 in the hash chain which is for the prior message N-1, and which includes the OTS public key for message N.

[0022] The techniques therefore permit even lightweight devices such as IoT devices to provide for authentication of their digitally signed messages in a way that is resistant to quantum computing attacks. Because only a single OTS public key may have to be created in advance, the techniques are computationally inexpensive and can be quickly performed by devices having lesser computing power (i.e., slower processors). The techniques further do not require large amounts of storage space, since a sending device only has to store a single OTS public key at any given time—the key that the device most recently generated and that will be used to authenticate the next message that is sent.

[0023] FIG. 1A shows an example hash chain **100** used to authenticate messages sent by a sending device to a receiving device. The hash chain **100** may be implemented as a blockchain such that it is managed by a peer-to-peer (P2P) computer network for use as a distributed ledger, where nodes collectively adhere to a consensus algorithm protocol to add and validate blocks. The hash chain **100** includes an origin block **102**, which is the root block of the hash chain **100** and thus the first block in the hash chain **100**. Blocks **104A**, **104B**, and **104C** of the hash chain **100** are also depicted, and are collectively referred to as the blocks **104**.

[0024] The block **104A** is linked to and added after the origin block **102**; the block **104B** is linked to and added after the block **104A**; and the block **104C** is linked to and added after the block **104B**. The hash chain **100** can ultimately include an indeterminate number of additional blocks **104**. In general, each new block **104** is linked to and added after the most recently added block **104** of the hash chain **100**. In the depicted example of FIG. 1A, the next block that is added to the hash chain **100** will be linked to the block **104C**, the block added after that will be linked to this block, and so on.

[0025] The origin block **102** includes an OTS public key **106A** corresponding to the OTS private key **108A** that is used to digitally sign the block **104A** when the block **104A** is added to the hash chain **100** after the origin block **102** has already been added. The origin block **102** itself may be digitally signed with a private key **110** corresponding to a public key **112**, resulting in the origin block **102** having a signature **114** by which the origin block **102** can be authenticated using the public key **112**. Authentication of the origin block **102** occurs in an out-of-band manner with respect to the hash chain **100**. That is, the origin block **102** is not authenticated using information stored within the hash chain **100**. For instance, the public key **112** may not be stored in the hash chain **100**. Authentication of the origin block **102** verifies that the origin block **102**—and thus the

public key **106A**—is associated with the sending device.

[0026] When the origin block **102** is generated, the OTS private key **108A** and its corresponding OTS public key **106A** are generated so that the OTS public key **106A** can be included in the origin block **102** before the origin block **102** is added to the hash chain **100**. The block **104A**, which as noted is added to the hash chain **100** after the origin block **102**, corresponds to an individual message **105A** from the sending device to a receiving device. The private key **108A** and the public key **106A** can be generated before the message **105A** is generated. Once the message **105A** has been generated, but before it has been sent, the block **104A** is generated and added to the hash chain **100**.

[0027] The block **104A** includes an OTS public key **106B** corresponding to the OTS private key **108B** that is used to digitally sign the (next) block **104B** when the block **104B** is added to the hash chain **100** after the block **104A** has already been added. The block **104A** includes a link hash **116A** of the origin block **102**, which links the block **104A** to the (prior) origin block **102** in the hash chain **100**. The link hash **116A** is a cryptographic hash of the origin block **102**.

[0028] The block **104A** includes a payload hash **118A** of the payload **120A** of the message **105A** to which the block **104A** corresponds. The payload **120A** is the data of the message **105A** that the sending device is to send to the receiving device in the message **105A**. The block **104A** may include a message index **121A** that identifies the message **105A**, and which may also be included in the message **105A** itself (such as in the payload **120A**). The message index **121A** may indicate the order of the message **105A** relative to other messages that the sending device sends to the receiving device. For example, each time a message is sent, its message index may be incremented for the next message.

[0029] The block **104A** is digitally signed with the OTS private key **108A** corresponding to the OTS public key **106A** included in the (prior) origin block **102**, such that the block **104A** has a signature **122A**. The message **105A** is thus authenticated by authenticating its corresponding block **104A** with the public key **106A**. Such authentication verifies that the block **104A** was signed with the private key **108A** corresponding to the public key **106A**, and is performed in an in-band manner with respect to the hash chain **100** in that the block **104A** is authenticated using information stored within the hash chain **100**. When the origin block **102** was previously authenticated (in an out-of-band manner with respect to the hash chain **100**), authentication of the block **104A** verifies that the corresponding message **105A** was sent by the sending device.

[0030] When the block **104A** is generated, the OTS private key **108B** and its corresponding OTS public key **106B** are generated so that the OTS public key **106B** can be included in the block **104A** before the block **104A** is added to the hash chain **100**. The block **104B**, which as noted is added to the hash chain **100** after the block **104A**, corresponds to an individual message **105B** from the sending device to a receiving device sent after the message **105A**. The private key **108B** and the public key **106B** can be generated before the message **105B** is generated. Once the message **105B** has been generated, but before it has been sent, the block **104B** is generated and added to the hash chain **100**.

[0031] The block **104B** includes an OTS public key **106C** corresponding to the OTS private key **108C** that is used to digitally sign the (next) block **104C** when the block **104C** is added to the hash chain **100** after the block **104B** has already been added. The block **104B** includes a link hash **116B** of the block **104A**, which links the block **104B** to the (prior) block **104A** in the hash chain **100**. The block **104B** includes a payload hash **118B** of the payload **120B** of the message **105B** to which the block **104B** corresponds. The block **104B** may include a message index **121B** that identifies the message **105B**, and which may also be included in the message **105B** itself.

[0032] The block **104B** is digitally signed with the OTS private key **108B** corresponding to the OTS public key **106B** included in the (prior) block **104A**, such that the block **104B** has a signature **122B**. The message **105B** is thus authenticated by authenticating its corresponding block **104B** with the public key **106B**. Such authentication verifies that the block **104B** was signed with the private

key **108B** corresponding to the public key **106B**, and is performed in an in-band manner with respect to the hash chain **100**. When both the origin block **102** and the block **104A** were previously authenticated, authentication of the block **104B** verifies that the corresponding message **105B** was sent by the sending device.

[0033] When the block **104B** is generated, the OTS private key **108C** and its corresponding OTS public key **106C** are generated so that the OTS public key **106C** can be included in the block **104B** before the block **104B** is added to the hash chain **100**. The block **104C**, which as noted is added to the hash chain **100** after the block **104B**, corresponds to an individual message **105C** from the sending device to a receiving device sent after the message **105B**. The private key **108C** and the public key **106C** can be generated before the message **105C** is generated. Once the message **105C** has been generated, but before it has been sent, the block **104C** is generated and added to the hash chain **100**.

[0034] The block **104C** includes an OTS public key **106D** corresponding to the OTS private key **108D** that is used to digitally sign the next block (not depicted in FIG. 1A) when that block is added to the hash chain **100** after the block **104C** has already been added. The block **104C** includes a link hash **116C** of the block **104B**, which links the block **104C** to the (prior) block **104B** in the hash chain **100**. The block **104C** includes a payload hash **118C** of the payload **120C** of the message **105C** to which the block **104C** corresponds. The block **104C** may include a message index **121C** that identifies the message **105C**, and which may also be included in the message **105C** itself.

[0035] The block **104C** is digitally signed with the OTS private key **108C** corresponding to the OTS public key **106C** included in the (prior) block **104B**, such that the block **104C** has a signature **122C**. The message **105C** is thus authenticated by authenticating its corresponding block **104C** with the public key **106C**. Such authentication verifies that the block **104C** was signed with the private key **108C** corresponding to the public key **106C**, and is performed in an in-band manner with respect to the hash chain **100**. When all prior blocks—i.e., the origin block **102** and the blocks **104A** and **104B**—were previously authenticated, authentication of the block **104C** verifies that the corresponding message **105C** was sent by the sending device.

[0036] When the block **104C** is generated, the OTS private key **108D** and its corresponding OTS public key **106D** are generated so that the OTS public key **106D** can be included in the block **104C** before the block **104C** is added to the hash chain **100**. The next block added to the hash chain **100** after the block **104C** corresponds to the next individual message to be sent from the sending device to a receiving device after the message **105C**. The private key **108D** and the public key **106D** can be generated before this next message is generated. Once this message has been generated, but before it has been sent, a corresponding block is generated and added to the hash chain **100**.

[0037] FIG. 1B more generally shows an example block **154** of the hash chain **100** other than the origin block **102**. Each of the blocks **104A**, **104B**, and **104C** is an instance of the block **154**. The block **154** is for message N, and thus can be referred to as block N. The block **154** includes an OTS public key **157A** used to authenticate the next block N+1, and therefore can be referred to as the public key N+1. The block **154** includes a link hash **166** of the prior block N-1, and a payload hash **168** of the payload of message N. The block **154** can include a message index **171** that identifies the message N, which may be referred to as the index N and which may also be included in the message N.

[0038] The block **154** is digitally signed with an OTS private key **158**, which can be referred to as the private key N that corresponds to the public key N included in the prior block N-1, such that the block **154** has a signature **172**. The block N is authenticated with the public key N included in the prior block N-1, which verifies that the block N was signed with the corresponding private key N. This authentication is performed in an in-band manner with respect to the hash chain **100**. When the origin block **102** and every prior non-origin block M ($0 < M < N$) in the hash chain **100** were previously authenticated, authentication of block N verifies that the message N was sent by the sending device.

[0039] FIG. 2 shows an example method **200** for sending messages that can be authenticated using the hash chain **100**. The method **200** is performed by the sending device. The sending device can set $N=1$ (**202**), where N may be the message index **171**. The sending device generates the cryptographic key pair $N=1$ for message $N=1$ that may not yet have been generated (**204**). Specifically, the sending device generates an OTS private key $N=1$ and its corresponding OTS public key $N=1$, such as in accordance with the LM-OTS scheme.

[0040] The sending device generates the origin block **102** for the hash chain **100** (**206**), such that the origin block **102** includes the OTS public key $N=1$. The sending device digitally signs the origin block **102** with a private key **110** having a corresponding public key **112** that is not stored in the hash chain **100** (**208**). The sending device creates the hash chain **100** with the origin block **102** as block $N=0$ (**210**). The origin block **102** is thus added to the hash chain **100**.

[0041] When message N is to be sent (**212**)—i.e., once message N has been generated and is ready to be sent to the receiving device—the sending device generates the payload hash **168** of the payload of message N (**214**), and generates the link hash **168** of the prior block $N-1$ (**216**). The sending device generates the cryptographic key pair $N+1$ for the next message $N+1$ that may not yet have been generated (**218**). The sending device thus generates the OTS private key $N+1$ and its corresponding OTS public key $N+1$.

[0042] The sending device then generates block N for message N (**220**), such that block N includes the payload hash **168** of the payload of message N , the link hash **166** of the prior block $N-1$, and the OTS public key $N+1$ for the next message $N+1$. Block N may also include the message index N . The sending device digitally signs block N with OTS private key N (**222**), such that block N includes the signature **172**, and adds digitally signed block N to the hash chain **100** (**224**). N is incremented by one (**226**), and the method **200** is repeated at **212** when the next message is to be sent.

[0043] FIG. 3 shows an example method **300** for receiving and authenticating messages using the hash chain **100**. The method **300** is performed by the receiving device. The receiving device can set $N=1$ (**302**), where N may be the message index as separately maintained at the receiving device. The receiving device retrieves the origin block **102** from the hash chain **100** (**304**), and authenticates the origin block **102** in an out-of-band manner with respect to the hash chain **100** (**306**). Authentication of the origin block **102** verifies that the origin block **102**—and thus public key $N=1$ included in the origin block **102**—is associated with the sending device.

[0044] For instance, the receiving device may receive the public key **112** used to authenticate the origin block **102** directly from the sending device, or from a trusted third party that vouches for the public key **112** as being associated with the sending device. In this manner, the public key **112** is trusted and is known to be associated with the sending device, such that authentication of the origin block **102** using the public key **112** verifies that the origin block **102** is itself associated with the sending device.

[0045] The receiving device receives message N (**308**), and in response retrieves from the block chain **100** block N that corresponds to message N (which includes public key $N+1$ for the next message $N+1$) (**310**). The receiving device can retrieve block N in one implementation by retrieving from the block chain **100** the block that has a message index **171** equal to N . The receiving device authenticates block N using public key N in block $N-1$ that was previously retrieved and authenticated (**312**). When $N=1$, block $N-1$ is the origin block **102**; when $N>1$, block $N-1$ is the immediately prior non-origin block. The receiving device authenticates block N in that it verifies that the signature **172** of block N has been signed with the private key N corresponding to the public key N .

[0046] Authentication may also include verifying that the link hash **166** included in block N refers to prior block $N-1$ that was previously retrieved and/or that the block that is referred to by the link hash **166** includes the public key N that was previously retrieved when the previously received message $N-1$ was authenticated. Authentication may further include verifying N as maintained at

the receiving device matches the message index of the message that has been received.

Authentication of block N may be considered successful only if prior authentication of the origin block **102** (as well as every prior non-origin block when $N > 1$) was successful. That is, block N is authenticated upon having successfully authenticated the origin block **102** (and every prior non-origin block when $N > 1$).

[0047] The receiving device further calculates the payload hash of the payload of received message N (**314**). (The payload hash may be calculated before retrieving block N, such that block N is retrieved by retrieving from the hash chain **100** the block that has a matching payload hash **168**, in a different implementation as compared to that noted above.) When the calculated payload hash matches the payload hash included in block N—and upon block N itself having been successfully authenticated—the receiving device authenticates that message N has been sent by the sending device. Therefore, the receiving device can perform an action based on received message N (i.e., based on the payload of received message N) (**316**).

[0048] In other words, because the receiving device has authenticated that message N has been sent by the sending device, the receiving device can act on the message. For example, if the payload of message N includes an action that the receiving device is to perform, the receiving device performs the action. If the payload of message N includes information regarding the sending device or other information, the receiving device can consider this information authentic and store and/or otherwise act on the information. N is incremented by one (**318**), and the method **300** is repeated at **308** when the next message is received.

[0049] In the examples of FIGS. **1A**, **1B**, **2**, and **3** that have been described, the origin block **102** does not correspond to a message **105**. However, in another implementation, the origin block **102** can correspond to a message **105**. That is, the origin block **102** is the same as any block **104** except that it does not link to a prior block (since it is the first block in the hash chain **100**), and is authenticated in an out-of-band manner instead of an in-band manner.

[0050] Furthermore, the example method **300** of FIG. **3** has been described on the assumption that every message sent by the sending device is received by the receiving device, where the receiving device authenticates the messages in the order in which the sending device sent the messages. So long as the receiving device receives every message sent by the sending device, the hash chain **100** can be used for an indeterminate number of messages. This means that the receiving device only has to perform authentication in an out-of-band manner once, as to the origin block **102**.

[0051] However, if a message sent by the sending device is not received by the receiving device, then the hash chain **100** can no longer be used for subsequently sent messages that are received by the receiving device. Rather, a new hash chain **100**, with a new origin block **102**, has to be created, and the receiving device has to perform out-of-band authentication again. In general, when the receiving device receives a message that is not consecutive to the immediately prior message sent by the sending device, a new hash chain **100** has to be created and a new origin block **102** added to that hash chain **100**, followed by out-of-band authentication of the new origin block **102**.

[0052] Furthermore, when the receiving device receives a message that is not consecutive to the immediately prior message sent by the sending device, authentication of that message will fail at the receiving device. The receiving device may thus signal to the sending device a new hash chain **100** with a new origin block **102** has to be created.

[0053] For example, the receiving device may receive and successfully

[0054] authenticate message **105A**. However, the next message that the receiving device receives may be the message **105C**. That is, the receiving device may have failed to receive the message **105B** that the sending device sent after the message **105A** and before the message **105C**. In this case, authentication of the message **105C** will fail at the receiving device. The hash chain **100** can no longer be used for authentication of messages from the sending device, and a new hash chain **100** with a new origin block **102** is created, followed by out-of-band authentication of the new origin block **102**.

[0055] FIG. 4, however, shows the non-origin block **154** in an example implementation that permits the hash chain **100** to be used even when the receiving device has failed to receive up to $M-1$ consecutive messages sent by the sending device. The block **154** is for message N and can be referred to as block N . As has been described in reference to FIG. 1B, the block **154** includes a link hash **166** of the prior block $N-1$ and a payload hash **168** of the payload of message N , and may further include a message index **171**. As has also been described, the block **154** is digitally signed with an OTS private key **158**, which can be referred to as the private key N that corresponds to the public key N included in the prior block $N-1$, such that the block **154** has a signature **172**.

[0056] The block **154** in the example of FIG. 4 includes M OTS public keys **157** respectively corresponding to the next M messages that will be sent. The OTS public key **157A** is the public key $N+1$ used to authenticate the next block $N+1$ for the next message $N+1$. The public key $N+2$ is for block $N+2$ and thus for message $N+2$ that will be sent after message $N+1$, and so on, where the public key $N+M$ is for block $N+M$ and thus for message $N+M$ that will be sent after message $N+M-1$. In this implementation, the origin block **102** includes the public keys $N=1$ through $N=M$.

[0057] If after receiving message N the receiving device fails to receive message $N+1$, $N+2$, $\dots$, $N+M-1$, the receiving device will still be able to successfully authenticate message $N+M$ in an in-band manner if that message is received. This is because block N includes public key $N+M$ corresponding to the private key $N+M$ that the sending device used to digitally sign block $N+M$, such that the receiving device can successfully authenticate block $N+M$. A new hash chain **100** with a new origin block **102** does not have to be created.

[0058] This implementation can be appropriately used to add a degree of resilience to the hash chain **100** when it is not expected that the receiving device will fail to receive more than a relatively small number $M-1$ of consecutively sent messages. However, in scenarios in which it is expected that the receiving device may fail to receive a relatively large number $M-1$ of consecutively sent messages, having to include such a large number M of OTS public keys **157** in each block **154** may be deemed untenable. For instance, the sending device will have to initially create the first M public keys **157** and their M corresponding private keys before sending the first message $N=1$, and may have to store at least the M corresponding private keys at all times. The sending device may also store the M public keys **157** at all times, or generate them as needed from the M corresponding private keys.

[0059] FIG. 5A, however, shows an example implementation that permits the hash chain **100** to be used even when the receiving device has failed to receive up to $M-1$ consecutively sent messages, and which may be appropriate for even a relatively large value of M . The implementation leverages Merkle trees, or another type of cryptographic binary tree, per the LMS, XMSS, or another such scheme. Specifically depicted are the first three Merkle trees **500A**, **500B**, and **500C**, which are collectively referred to as the Merkle trees **500**.

[0060] Each Merkle tree **500** includes a root hash **502**, (intermediate-level) child hashes **504** (e.g., the child hashes **504A** and **504B** of the tree **500A**), and leaf hashes **506** (e.g., the leaf hashes **506A**, **506B**, **506C**, and **506D** of the tree **500A**). The root hash **502** of a Merkle tree **500** corresponds to a root public key **503**, and the leaf hashes **506** respectively correspond to OTS public keys **507**. The public keys **507** themselves correspond to OTS private keys **508** used to digitally sign blocks **510** that are added to the hash chain **100** when corresponding messages **512** are sent.

[0061] The leaf hashes **506** are also technically child hashes, but because they do not themselves have child hashes are not labeled as such in the figure. Since the tree **500** is a binary tree, every hash other than the leaf hashes **506** has two child hashes. In the example, each Merkle tree **500** has a height $h=2$, which means that the tree **500** has two levels of hashes below the root hash **502**, the level of the child hashes **504** and the level of leaf hashes **506**. More generally, each Merkle tree **500** has a height $h>1$, such that the tree **500** has $h-1$ intermediate levels of child hashes **504** and one level of leaf hashes **506**.

[0062] A Merkle tree **500** is created, such as per the LMS or XMSS scheme, as follows. For a

specified height h , 2.sup.h OTS private keys **508** and 2.sup.h corresponding OTS public keys **507** are generated. The leaf hashes **506** correspond to their respective public keys **507** in that they are hashes of the public keys **507** or are the public keys **507**. Each child hash **504** is in general a hash of the hashes of its two child hashes. In the example, each child hash **504** is specifically a hash of its two leaf hashes **506**. For the tree **500A** specifically, the child hash **504A** is a hash of the leaf hashes **506A** and **506B**, and the child hash **504B** is a hash of the leaf hashes **506C** and **506D**.

[0063] The root hash **502** is similarly a hash of its two child hashes **504**. For the Merkle tree **500A** specifically, the root hash **502** is a hash of the child hashes **504A** and **504B**. The root hash **502** corresponds to the root public key **503**. As one example, the root public key **503** may be generated so that the root hash **502** is the hash of the root public key **503** (as well as being the hash of the leaf hashes **506A** and **506B**). As another example, the root public key **503** may be the root hash **502** (i.e., the root public key **503** can be the hash of the leaf hashes **506A** and **506B**). As a third example, the root hash **502** may include the root public key **503** as well as other information.

[0064] The block **510** for a message **512** is digitally signed with its corresponding OTS private key **508**. For the receiving device to authenticate the block **510**, and thus to authenticate the message **512**, the corresponding public key **507**, the authentication path from the leaf hash **506** for the public key **507** to the root hash **502** in the Merkle tree **500** in question, and the root public key **503** of that tree **500** are needed. The public key **507** is calculated by the receiving device from the signature **172** of the block **510**. The authentication path is included in the block **510**, whereas the root public key **503** is already known (i.e., has already been received or retrieved by) to the receiving device.

[0065] The receiving device is provided with the authentication path from the leaf hash **506** for the public key **507** to the root hash **502** (via inclusion of the authentication path in the block **510** itself) because the receiving device is not provided with and therefore does not have the Merkle tree **500** itself. The authentication path includes the information sufficient for the receiving device to calculate the root hash **502** from the public key **507** for the message **512**. The receiving device authenticates the block **510** by determining that this calculated root hash **502** corresponds to the root public key **503** that is already known by the receiving device.

[0066] For example, the authentication path for the leaf hash **506A** can include the leaf hash **506B** and the child hash **504B**, which are sufficient for the receiving device to calculate the root hash **502** from the public key **507** that the receiving device calculated from the signature **172**, and which if the message **512** indeed has been sent by the sending device corresponds to the leaf hash **506A**. The receiving device calculates the leaf hash **506A** from the calculated public key **507**, calculates the child hash **504A** from the calculated leaf hash **506A** and the provided leaf hash **506B**, and calculates the root hash **502** from the calculated child hash **504A** and the provided child hash **504B**.

[0067] The receiving device then determines whether the calculated root hash **502** matches the root public key **503** that it already knows. For example, if the root public key **503** is itself the root hash **502**, then the calculated root hash **502** should be the root public key **503** that the receiving device already knows. In this case, if the calculated root hash **502** is not the known root public key **503**, then authentication of the message is unsuccessful; that is, the message is not authenticated as having been sent from the sending device.

[0068] As another example, the authentication path for the leaf hash **506D** can include the leaf hash **506C** and the child hash **504A**, which are sufficient for the receiving device to calculate the root hash **502** from the public key **507** corresponding to the leaf hash **506D**. The receiving device first calculates the public key **507** from the signature **172**. The receiving device then calculates the leaf hash **506D** from the provided public key **507**, calculates the child hash **504B** from the calculated leaf hash **506D** and the provided leaf hash **506C**, and calculates the root hash **502** from the calculated child hash **504B** and the provided child hash **504A**. If the calculated root hash **502** matches the known root public key **503**, then the message has been successfully authenticated as having been sent from the sending device.

[0069] FIG. 5B shows the non-origin block **154** in this example implementation. Each block **510** is

an instance of the non-origin block **154**. The block **154** is for message N and therefore can be referred to as block N. As has been described in reference to FIG. **1B**, the block **154** includes a link hash **166** of the prior block N-1 and a payload hash **168** of the payload of message N, and may further include a message index **171**. The block **154** is digitally signed with an OTS private key **558**, which can be referred to as the private key N, such that the block **154** has a signature **172**. The OTS public key corresponding to the OTS private key **558** can be referred to as the public key N.

[0070] The block **154** includes the authentication path **559** from the leaf hash **506** corresponding to the public key N to the root hash **502** corresponding to the root public key **503** in the Merkle tree **500**. The block **154** includes the root public key **157** of the next Merkle tree **500**, in lieu of the public key **157A** for the next block N+1 as in FIG. **1B**, and in lieu of the public keys **157** for the next M blocks as in FIG. **4**. The origin block **102** in this implementation includes the root public key **157** of the first Merkle tree **500A** in lieu of the public key **507** for the first block N=1.

[0071] The blocks **510** for the first four messages **512** therefore each include the root public key **503** for the second Merkle tree **500B**. The blocks **510** for the next four messages **512** after that each include the root public key **503** for the third Merkle tree **500C**. The root public key **503** for the first Merkle tree **500A** is included in the origin block **102**.

[0072] After having authenticated the origin block **102** in an out-of-band manner, the receiving device can authenticate any received message **512** corresponding to the first Merkle tree **500A** in an in-band manner. The receiving device is able to authenticate any such message **512** regardless of whether it has received any other such message **512**. That is, authentication of a received message **512** corresponding to the tree **500A** does not depend on having authenticated or even having received any other message **512** which corresponds to the tree **500A**.

[0073] Similarly, after having authenticated the origin block **102** in an out-of-band manner and at least one received message **512** corresponding to the first Merkle tree **500A** in an in-band manner, the receiving device can authenticate any received message **512** corresponding to the second Merkle tree **500B** in the in-band manner. The receiving device is able to authenticate any such message **512** regardless of whether it has received any other such message **512**. Authentication of a received message **512** corresponding to the tree **500B** does not depend on having authenticated or even having received any other message **512** corresponding to the tree **500B**.

[0074] In general, for authentication of a received message **512** corresponding to a given Merkle tree **500**, the origin block **102** and only a single message **512** corresponding to every prior Merkle tree **500** (if any) has to have been previously authenticated. If no message **512** for a given Merkle tree **500** is received, however, a new hash chain **100** has to be created with an origin block **102** including the root public key **503** of the next Merkle tree **500** before messages **512** corresponding to the next Merkle tree **500** can be authenticated.

[0075] This implementation also adds a degree of resilience to the hash chain **100**. Prior to sending the first message **512**, the sending device just has to create the first two Merkle trees **500A** and **500B** so that the root public key **503** for the first tree **500A** can be included in the origin block **102** and so that the root public key **503** for the second tree **500B** can be included in the blocks **510** for the first 2.sup.h messages **512**. The sending device therefore just has to generate 2×2.sup.h OTS private keys **508** and their corresponding public OTS public keys **507** before sending the first message **512**.

[0076] Once the first 2.sup.h messages **512** have been sent, the sending device then has to create the third Merkle tree **500C** so that the root public key **503** for the third tree **500C** can be included in the blocks **510** for the next 2.sup.h messages **512**. In general, after every 2.sup.h messages **512** are sent, the sending device has to create another Merkle tree **500**. Therefore, after every 2.sup.h messages **512** are sent, the sending device has to generate 2.sup.h OTS private keys **508** and their corresponding OTS public keys **507** before sending any of the next 2.sup.h messages **512**.

[0077] FIG. **6** shows an example system **600** including a sending device **602A** and a receiving device **602B** communicatively connected to one another via a network **603**. Each of the devices

602A and **602B** can be a computing device or another type of electronic device. Examples of computing devices include desktop, laptop, notebook, and server computers, as well as other computing devices, including smartphones, tablet computing devices, and so on. Other types of electronic devices include networking devices such as routers, firewalls, access points, and so on, as well as printing and/or image devices such as printers, scanners, and all-in-one (AIO) devices that have at least printing functionality and scanning functionality. At least one of the devices **602A** and **602B** may be a lightweight device such as an IoT device. The network **603** may be or include the Internet and/or other types of networks.

[0078] The sending device **602A** and the receiving device **602B** respectively include processors **604A** and **604B** and memory **606A** and **606B**, the latter which are each more generally a non-transitory computer-readable data storage medium. The memory **606A** stores program code **608A** executable by the processor **604A** of the sending device **602A** to perform processing to send messages to the receiving device **602B** that can be authenticated using the hash chain **100** as has been described. The memory **606B** stores program code **608B** executable by the processor **604B** of the receiving device **602B** to perform processing to receive and authenticate the messages using the hash chain **100** as has been described.

[0079] Techniques have been described herein for using a hash chain of blocks of OTS public keys to authenticate messages in a way that is resistant to quantum computing attacks but that can still be performed by even lightweight devices such as IoT devices. The techniques are computationally inexpensive in that just a single OTS public key may have to be created in advance. The techniques further do not require large amounts of storage space, and just a single OTS public key may have to be stored at any given time.

Claims

1. A method performed by a sending device and comprising: generating a block to be added to a hash chain, the block for a message to be sent to a receiving device, the block including: a link hash of a prior block previously added to the hash chain, the prior block including a public key corresponding to a private key to be used to digitally sign the block; a payload hash of a payload of the message; and a next public key corresponding to a next private key to be used to digitally sign a next block to be added to the hash chain after the block, the next block for a next message to be sent to the receiving device after the message; digitally signing the block with the private key; adding the block to the hash chain, the link hash included in the block linking the block to the prior block in the hash chain; and sending the message to the receiving device.
2. The method of claim 1, wherein the public key is a one-time signature (OTS) public key such that the private key to which the public key corresponds is used only to digitally sign the block for the message and not to digitally sign any other block for any other message.
3. The method of claim 1, wherein the link hash is generated using a cryptographic hash function.
4. The method of claim 1, wherein the block further includes a message index identifying the message.
5. The method of claim 1, wherein digitally signing the block comprises generating a digital signature of at least the link hash, the payload hash, and the next public key, and wherein the method further comprises adding the digital signature to the block before adding the block to the hash chain.
6. The method of claim 1, wherein the block is authenticated by the receiving device in an in-band manner with respect to the hash chain, using the public key included in the prior block.
7. The method of claim 6, wherein the prior block is an origin block and is authenticated by the receiving device in an out-of-band manner with respect to the hash chain, and wherein authentication of the block is successful only if prior authentication of the origin block was successful.

8. The method of claim 6, wherein the prior block is not an origin block and is authenticated by the receiving device in the in-band manner with respect to the hash chain, wherein the origin block is authenticated by the receiving device in an out-of-band manner with respect to the hash chain, wherein authentication of the prior block is successful only if prior authentication of at least the origin block was successful, and wherein authentication of the block is successful only if prior authentication of at least the prior block and the origin block was successful.

9. The method of claim 1, wherein the next public key, the next private key, the next block, and the next message are respectively a first public key, a first private key, a first block, and a first message, wherein the block further includes a second public key corresponding to a second private key to be used to digitally sign a second block to be added to the hash chain after the first block, the second block for a second message to be sent to the receiving device after the first message, and wherein the block further includes a third public key corresponding to a third private key to be used to digitally sign a third block to be added to the hash chain after the second block, the third block for a third message to be sent to the receiving device after the second message.

10. The method of claim 9, wherein inclusion of the second public key permits the receiving device to authenticate the second message in an in-band manner with respect to the hash chain even if the receiving device does not receive the first message, and wherein inclusion of the third public key in the block permits the receiving device to authenticate the third message in the in-band manner with respect to the hash chain even if the receiving device does not receive the first message and the second message.

11. The method of claim 1, wherein the private key with which the block is digitally signed corresponds to a leaf public key that corresponds to a leaf hash of a binary tree of hashes including a root hash that corresponds to a root public key and a plurality of other leaf hashes that respectively correspond to other leaf public keys, wherein the public key included in the prior block is the root public key, wherein the block further includes an authentication path from the leaf hash to the root hash in the binary tree, wherein the next public key included in the block is a next root public key corresponding to a root hash of a next binary tree.

12. The method of claim 11, wherein the block is authenticated by the receiving device in an in-band manner with respect to the hash chain, using the root public key included in the prior block, the authentication path included in the block, and the leaf public key, which the receiving device computes, and wherein inclusion of the root public key in the prior block permits the receiving device to authenticate any subsequently received message corresponding to another leaf hash in the binary tree regardless of whether any other message corresponding to any other leaf hash in the binary tree has been received.

13. A sending device comprising: a processor; and a memory storing program code executable by the processor to: generate a first private key to be used to digitally sign a first block to be generated for a first message to be sent to a receiving device, and a first public key corresponding to the first private key; generate an origin block to be added to a hash chain, the origin block including the first public key; digitally sign the origin block; add the digitally signed origin block to the hash chain; generate a second private key to be used to digitally sign a second block for a second message to be sent to the receiving device, and a second public key corresponding to the second private key; generate a first block to be added to the hash chain, the first block for the first message including a link hash of the origin block, a payload hash of a payload of the first message, and the second public key; digitally sign the first block with the first private key; add the digitally signed first block to the hash chain, the link hash of the origin block included in the first block linking the first block to the origin block in the hash chain; and send the first message to the receiving device.

14. The sending device of claim 13, wherein the program code is executable by the processor to further: generate a third private key to be used to digitally sign a third block for a third message to be sent to the receiving device, and a third public key corresponding to the third private key; generate a second block to be added to the hash chain, the second block for the second message

including a link hash of the first block, a payload hash of a payload of the second message, and the third public key; digitally sign the second block with the second private key; add the second block to the hash chain, the link hash of the first block included in the second block linking the second block to the first block in the hash chain; and send the second message to the receiving device.

15. The sending device of claim 13, wherein the program code is executable by the processor to further: generate a third private key to be used to digitally sign a third block for a third message to be sent to the receiving device, and a third public key corresponding to the third private key, wherein the first block further includes the third public key; generate a fourth private key to be used to digitally sign a fourth block for a fourth message to be sent to the receiving device, and a fourth public key corresponding to the fourth private key; generate a second block to be added to the hash chain, the second block for the second message including a link hash of the first block, a payload hash of a payload of the second message, the third public key, and the fourth public key; digitally sign the second block with the second private key; add the digitally signed second block to the hash chain, the link hash of the first block included in the second block linking the second block to the first block in the hash chain; and send the second message to the receiving device.

16. The sending device of claim 13, wherein the first public key and the second public key are leaf public keys respectively corresponding to leaf hashes of a binary tree including a root hash corresponding to a root public key, wherein the first block further includes an authentication path from the leaf hash corresponding to the first public key to the root hash corresponding to the root public key, wherein the origin block includes, in lieu of the first public key, the root public key, and wherein the first block includes, in lieu of the second public key, a next root public key corresponding to a root hash of a next binary tree.

17. A non-transitory computer-readable data storage medium storing program code executable by a receiving device to perform processing comprising: receiving a first message from a sending device; retrieving a digitally signed first block of a hash chain, the first block corresponding to the first message and digitally signed with a first private key corresponding to a first public key, the first block including a link hash of an origin block of the hash chain, a payload hash of a payload of the first message, and a second public key for a second message from the sending device; retrieving the origin block of the hash chain using the link hash, the origin block including the first public key; authenticating the origin block in an out-of-band manner with respect to the hash chain; upon successfully authenticating the origin block, authenticating the first block using the first public key included in the origin block; calculating the payload hash of the payload of the first message received from the sending device; and upon successfully authenticating the first block and when the calculated payload hash matches the payload hash included in the first block, performing a first action based on the first message.

18. The non-transitory computer-readable data storage medium of claim 17, wherein the processing further comprises: receiving the second message from the sending device; retrieving a digitally signed second block of the hash chain, the second block corresponding to the second message and digitally signed with a second private key corresponding to the second public key, the second block including a link hash of the first block of the hash chain, a payload hash of a payload of the second message, and a third public key for a third message from the sending device; authenticating the second block using the second public key included in the first block; calculating the payload hash of the payload of the second message received from the sending device; and upon successfully authenticating the second block and when the calculated payload hash of the payload of the second message matches the payload hash of the payload of the second message included in the second block, performing a second action based on the second message.

19. The non-transitory computer-readable data storage medium of claim 18, wherein the first block further includes the third public key for the third message, and wherein the second block further includes a fourth public key for a fourth message from the sending device.

20. The non-transitory computer-readable data storage medium of claim 18, wherein the first,

second, and third public keys are leaf public keys respectively corresponding to leaf hashes of a binary tree including a root hash corresponding to a root public key, wherein the first block includes an authentication path from the leaf hash corresponding to the first public key to the root hash corresponding to the root public key, wherein the second block includes an authentication path from the leaf hash corresponding to the second public key to the root hash corresponding to the root public key, wherein the origin block includes, in lieu of the first public key, the root public key, wherein the first block includes, in lieu of the second public key, a next root public key corresponding to a root hash of a next binary tree, and wherein the second block includes, in lieu of the third public key, the next root public key.
